

SBC 201 Chapter 23 Woods — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 4–10 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 23, source file Reference\SBC 201 2024\source_reference\Chapter_23 – WOODS.txt.
- **Extraction audit:** Skill extract. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold SI values, building-language triggers, named exceptions, check-specific actions). Internal inventory: **286** independently checkable numeric records (**148** Verified, **138** Verify source). Flattened fastening, sheathing, diaphragm and shear-wall tables are listed in the register and are not design-release values.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination. It is not a stamped compliance statement, structural wood design, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from Section 603.1, Table 601, Chapter 6 construction-type hours, Chapter 8 flame-spread classes, Chapter 16 loads, Chapter 17 special inspections, Section 1202 ventilation rates, Sections 2111 / 2113, SBC 301, SBC 303, AWC NDS, AWC SDPWS, AWC WFCM, AWC PWF, TPI 1, ICC 400, commentary examples, figure notes, bibliography years, or the existing chapter summary. Where Chapter 23 sends the user elsewhere, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications.
2. The exact Riyadh AHJ/permit pathway, project stage, fire-strategy status and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. Type of construction is unconfirmed. A Group R-2 high-rise is typically Type I (or II). This chapter sizes and qualifies wood members; it does not assign construction type. Type I/II wood use is limited by **603.1**, which is not reprinted here.
4. Automatic sprinkler protection is not selected. Chapter 23 does not branch on NFPA 13 versus 13R except Table 2304.11 note b (heavy-timber roof-member width under an approved sprinkler), which is **Verify source** / Conditional Type IV.
5. Mixed-use podium, storey count, wood sleepers, wood balcony framing, FRTW partitions, millwork and mass timber are unconfirmed.
6. Conventional light-frame Section **2308** is charged in this extract as **deemed inapplicable to Saudi Arabia**. WFCM (2309) is limited to Risk Category I or II buildings inside the WFCM’s own size and load limits and is not a high-rise wood-frame method.
7. Appended fastening, sheathing-span, diaphragm and shear-wall tables are concatenated OCR. No reconstructed table cell is adopted as a design-release value.

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, wood product, construction type, load, sprinkler branch or exception exists.
Not typical	Unrelated occupancy-only rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 23 points to another section/code/standard, or the project/AHJ basis must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 23 source text.
Verify source	OCR, flattened table, page-split, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 23 application	Required project action
Type of construction	Unconfirmed; high-rise R-2 typically Type I or II	2303.2.9 sends Type I/II FRTW to 603.1 . Light-frame wood is not a Type I shaft system. Type IV mass timber would open 2304.11	Freeze Type I-A / I-B (or other) on the code data sheet before specifying any wood
Wood products actually used	Unconfirmed: FRTW partitions, blocking, nailers, doors/frames, paneling, sleepers, balcony framing, or none	2301.1 only governs wood members and fasteners that exist. 2304 construction details apply when wood is used	List every wood item on the finish, carpentry and fire-stopping schedules; if none, most Chapter 23 rows stay unused
FRTW vs millwork vs blocking	Unconfirmed	FRTW must meet 2303.2 identity tests. Doors, frames, trim and blocking are Type I combustibles only where 603.1 lists them; those 603.1 millimetre and hour limits are not in this chapter	Separate FRTW, millwork and blocking on the specification; do not label paint or coating as FRTW (2303.2.2)
Type IV / mass timber	Unconfirmed; not assumed for this tower shaft	2304.11 and Table 2304.11 size heavy timber / CLT. Table 2304.11 body is OCR. Chapter 6 already sends Type IV member sizes here	If mass timber is selected, freeze IV-A / IV-B / IV-C and verify published Table 2304.11; otherwise keep 2304.11 Conditional
Mixed-use podium Type III / V	Unconfirmed	Would open wood-frame 2304 / 2305 / 2306 tables that are omitted from lead matrices as not typical for the Type I shaft	If a wood-framed podium is added, re-filter this chapter; do not apply fastening or diaphragm tables to the concrete tower
Sprinkler vs Table 2304.11 note b	Unconfirmed	Note b allows heavy-timber roof framing not less than 75 mm nominal width where protected by approved sprinklers under the roof deck. Flattened table body is not adopted	Do not reduce Type IV roof-member width from a reconstructed table; confirm the published note if Type IV is used
Termite / decay AHJ	Unconfirmed; Riyadh climate not a substitute for the code test	2304.12.2.7 applies only where termite hazard is known to be very heavy . 2304.12.2.3 Exception allows untreated sawn lumber where climate precludes decay — AHJ confirmation required	Ask the AHJ whether a very-heavy termite designation or a dry-climate untreated-sawn exception applies; do not assume either

Decision / gap	Current project basis	Why it controls Chapter 23 application	Required project action
Structural wood PE / NDS / SDPWS	Unconfirmed	2302.1, 2306.1 and 2307.1 send engineered wood to AWC NDS and SDPWS. This extract has no member stresses	Engage the structural engineer of record before sizing any load-carrying wood
NOC / fire strategy	Unconfirmed	FRTW listings, Type I combustible allowances and balcony wood cannot be concluded from Chapter 23 alone	Coordinate SCD NOC and the fire strategy before freezing FRTW partitions or wood balcony framing

4. Scope and design methods

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2301.1	Chapter wood scope	This chapter governs materials, design, construction and quality of wood members and their fasteners	Any wood member or wood fastener on the building	None stated	Direct	Put every wood work package on the drawing index; send load-carrying wood through 2302.1	Verified
2301.2	Nominal lumber sizes	Where this chapter specifies lumber dimensions, they are nominal unless specifically designated as actual dimensions	Lumber sizes stated in this chapter	Actual sizes govern design computations under 2304.2	Direct	Label millwork and blocking as nominal or actual on the schedule; do not mix the two	Verified
2302.1	Wood design method	Design of structural elements or systems constructed partially or wholly of wood or wood-based products shall use one of: ASD (2304, 2305, 2306); LRFD (2304, 2305, 2307); conventional light-frame (2304 and 2308); AWC WFCM (2309); or ICC 400 for log structures	Structural wood elements or systems	Partial design of a single element is recognised. Conventional and WFCM are not the Type I tower frame method	External verification	Name ASD or LRFD and the AWC standard on any structural-wood drawing; do not import NDS/SDPWS stresses here	Verified
2304.1–2304.2	General construction and actual sizes	Section 2304 applies to the design methods in 2302.1. Required member sizes shall be computed from net (actual) dimensions, not nominal sizes	Wood members designed or detailed under this chapter	None stated	Conditional	If any wood member carries load, size it on actual section; do not use nominal 2× dimensions in calcs	Verified
2305.1	Wood lateral system	Wood-frame shear walls or wood-frame diaphragms resisting wind, seismic or other lateral loads shall be designed and constructed in accordance with AWC SDPWS and the applicable provisions of 2305, 2306 and 2307	Wood-frame shear walls or diaphragms	Staple deflection equations 23-1 / 23-2 and Tables 2305.2(1)–(2) / 2306.2 / 2306.3 are wood-frame only and are flattened OCR	Conditional	If a wood podium LFERS is added, send it to SDPWS; do not use reconstructed staple-shear cells on the tower	Verified
2306.1 / 2307.1	ASD and LRFD standards	ASD wood elements shall comply with the standards in Table 2306.1 (AWC NDS and SDPWS named). LRFD wood elements shall comply with ANSI/AWC NDS and AWC SDPWS	Engineered wood using ASD or LRFD	Table 2306.1 is a standards list, not a size table. NDS/SDPWS values are not imported	External verification	Name NDS and SDPWS on the structural specification; do not copy allowable stresses into this matrix	Verified
2308.1	Conventional light-frame (KSA)	Conventional light-frame provisions are intended for that construction. Other methods are permitted with a satisfactory design. Because of its limited use it is deemed inapplicable to Saudi Arabia ; SBC 201 Section 2308 may be consulted if needed	Conventional light-frame wood buildings	Engineered ASD/LRFD remains available. Do not extract 2308 span or braced-wall tables	Direct	Do not use Section 2308 as a prescriptive wood-frame path on this KSA tower	Verified
2309.1	WFCM applicability	Structural design in accordance with AWC WFCM is permitted for buildings assigned to Risk Category I or II subject to the limitations of WFCM Section 1.1.3 and the load assumptions therein	Buildings inside WFCM size, configuration and load limits	Structural elements beyond those limits shall be designed in accordance with accepted engineering practice. WFCM values are not imported	Conditional	Do not apply WFCM dwelling tables to a high-rise shaft; if a low-rise wood ancillary building is added, confirm Risk Category and WFCM 1.1.3	Verified

5. Fire-retardant-treated wood identity and Type I/II pointer

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2303.2	FRTW flame-spread identity	FRTW is wood impregnated with chemicals by a pressure process or other means during manufacture that, tested to ASTM E84 or UL 723, has a listed flame spread index of 25 or less	Product specified as fire-retardant-treated wood	Surface paints, coatings and stains are not FRTW (2303.2.2)	Direct	Require an ASTM E84 / UL 723 listing of flame spread ≤ 25 on every FRTW submittal	Verified
2303.2	FRTW 20-minute flame front	The ASTM E84 or UL 723 test shall be continued for a 20-minute period and the flame front shall not progress more than 3.2 m beyond the burner centerline at any time	FRTW qualification test	Both the index 25 and the 3.2 m / 20-minute continuation apply together	Direct	Reject listings that omit the 20-minute flame-front result	Verified
2303.2.1	FRTW pressure process	Pressure impregnation shall be performed in closed vessels under pressures not less than 350 kPa	FRTW treated by a pressure process	Other-means manufacture is 2303.2.2, not a coating	Conditional	Confirm treating records show closed-vessel pressure ≥ 350 kPa	Verified
2303.2.2	Non-coating FRTW process	Other-means treatment shall be an integral part of manufacture and shall permanently protect all surfaces. Paints, coatings, stains or other surface treatments are not an approved method	Product claiming FRTW by coating or stain	Pressure-process products follow 2303.2.1	Direct	Prohibit intumescent paint or stain as a substitute for 2303.2 FRTW on Type I drawings	Verified
2303.2.3	FRTW panel test gap	Wood structural panels shall be tested with a ripped or cut longitudinal gap of 3.2 mm	FRTW wood structural panels	None stated	Conditional	Check the listing that panels were tested with a 3.2 mm gap	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2303.2.4	FRTW piece label	Each piece of FRTW lumber and wood structural panels shall be labeled with the approved-agency mark, treater, treatment name, species, flame-spread and smoke-developed index, drying method, and 2303.2.5–2303.2.8 standards. Weather/damp/wet use shall include the ASTM D2898 rain-test statement	FRTW lumber or panels on site	Grade marks under 2303.1.1 / 2303.1.5 remain in addition to the FRTW label	Direct	Require both the grade mark and the distinct FRTW label on delivered pieces	Verified
2303.2.5–2303.2.5.2	FRTW strength adjustment	Untreated design values shall be adjusted for FRTW by an approved investigation of temperature, humidity, treatment and redrying. Panels: ASTM D5516 and ASTM D6305; lumber: ASTM D5664 and ASTM D6841. Manufacturers shall publish panel loads/spans and lumber modification factors for service at temperatures not less than 27°C and for roof framing	FRTW used as a structural member	Adjustment factors live in the manufacturer’s published data, not in this chapter	External verification	Use only the treater’s published FRTW design values; do not use untreated NDS stresses	Verified
2303.2.6	Exterior FRTW identity	FRTW exposed to weather, damp or wet locations shall be identified as “ Exterior ” (no increase in the listed flame-spread index when subjected to ASTM D2898)	FRTW in weather, damp or wet locations	Interior-only FRTW is 2303.2.7	Conditional	Specify Exterior FRTW for any wet or weather-exposed piece	Verified
2303.2.7	Interior FRTW humidity test	Interior FRTW shall have moisture content not over 28 percent when tested to ASTM D3201 at 92-percent relative humidity, and shall be tested to 2303.2.5.1 or 2303.2.5.2. Type A interior FRTW shall be tested in accordance with this section	Interior FRTW	Keep the published 92-percent token intact	Direct	Require the ASTM D3201 28 percent / 92-percent RH report on interior FRTW	Verified
2303.2.8	FRTW moisture before use	FRTW shall be dried to 19 percent or less moisture for lumber and 15 percent or less for wood structural panels before use. KDAT kiln temperatures shall not exceed those used for the 2303.2.5 tests	FRTW installed in the building	None stated	Direct	Hold FRTW off the building until moisture tickets show lumber $\leq$ 19 percent and panels $\leq$ 15 percent	Verified
2303.2.9	Type I/II FRTW pointer	See Section 603.1 for limitations on FRTW in buildings of Type I or II construction	Type I or II building	603.1 hours, storey limits and the 6 m roof clearance are not in this chapter and are not imported	External verification	Take Type I/II FRTW locations from Section 603.1 / the Chapter 6 matrix; do not copy those numbers here	Verified

6. Product quality and identification

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2303.1	Listed wood products	Structural sawn lumber, end-jointed lumber, I-joists, glulam, wood structural panels, fiberboard and hardboard used structurally, particleboard, preservative-treated wood, logs, SCL, poles/piles, FRTW, hardwood plywood, trusses, joist hangers, nails and staples shall conform to this section	Those products used structurally	Particleboard outside this section must comply with 2306.3	Conditional	Name the 2303.1 standard on every structural-wood specification section that is actually used	Verified
2303.1.1–2303.1.1.1	Sawn-lumber grade mark	Load-supporting sawn lumber shall bear the grade mark of an approved DOC PS 20 (or equivalent) agency. A certificate of inspection may replace the mark for precut, remanufactured or rough-sawn lumber and for sizes larger than 75 mm nominal thickness	Load-supporting sawn lumber	End-jointed lumber in a fire-resistance-rated assembly shall include HRA on the grade mark (2303.1.1.2)	Conditional	Require grade marks on delivered lumber; accept a certificate only for the listed > 75 mm / remanufactured cases	Verified
2303.1.4–2303.1.5	CLT and panel identity	Cross-laminated timber shall be manufactured and identified to ANSI/APA PRG 320 . Wood structural panels used structurally shall conform to DOC PS 1, DOC PS 2 or ANSI/APA PRP 210 and be trademarked for grade, bond classification and Performance Category. Permanently outdoor-exposed panels shall be Exterior type, except underside-exposed roof sheathing may be Exposure 1	CLT or structural wood panels	Performance Category is the nominal panel thickness wherever this code references panel thickness	Conditional	Specify PRG 320 on any CLT; specify Exterior or Exposure 1 on weather-exposed panels	Verified
2303.1.7	Hardboard underlayment thickness	Hardboard underlayment shall meet the strength requirements of 5.5 mm or 6.5 mm service-class hardboard planed or sanded on one side to a uniform thickness of not less than 5.0 mm	Hardboard used as underlayment	Siding, prefinished paneling and other hardboard follow ANSI A135.6 / A135.5 / A135.4	Conditional	If hardboard underlayment is used, specify service class 5.5 mm or 6.5 mm and finished thickness $\geq$ 5.0 mm	Verified
2303.1.8.1	Particleboard underlayment thickness	Type PBU particleboard underlayment shall be not less than 6.5 mm thick and installed per Composite Panel Association instructions	Particleboard floor underlayment	Other particleboard uses only as 2303.1.8 / 2306.3 allow	Conditional	If PBU underlayment is used, specify thickness $\geq$ 6.5 mm	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2303.1.9–2303.1.9.2	Preservative-treated identity	Wood required by 2304.12 to be preservative-treated shall conform to AWPA U1 and M4 , bear an accredited quality mark, and, where used in enclosed locations that cannot readily dry, be at 19 percent moisture or less before being covered	Wood required to be preservative-treated by 2304.12	Permanent wood foundations go to SBC 303 ; those values are not imported	Conditional	Require AWPAs marks and ≤ 19 percent moisture before covering treated wood in drywall or flooring	Verified
2303.4.1.3	Long-span truss bracing PE	For trusses with clear spans 18 m or greater, a qualified registered design professional shall design temporary installation restraint and diagonal bracing and the PITMR / PITMDB	Wood trusses spanning 18 m or more	Shorter trusses still require 2303.4 drawings and TPI 1 / Chapter 17 as applicable	Conditional	If any wood truss reaches 18 m clear span, retain a PE for temporary and permanent bracing	Verified

7. Fasteners in treated wood

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2304.10.6	Treated-wood fastener charging	Fasteners, including nuts and washers, and connectors in contact with preservative-treated or fire-retardant-treated wood shall comply with 2304.10.6.1 through 2304.10.6.4. Zinc coating weights shall comply with ASTM A153 . Stainless driven fasteners shall comply with ASTM F1667	Fasteners touching PT or FRTW	Wood-foundation fastenings go to AWC PWF (2304.10.6.2); PWF values are not imported	Direct	Put corrosion-resistant fastener notes on every FRTW / PT carpentry detail	Verified
2304.10.6.1	Preservative-treated fasteners	Fasteners in contact with preservative-treated wood shall be hot-dipped zinc-coated galvanized steel, stainless steel, silicon bronze or copper. Staples shall be stainless steel. Other fasteners may be mechanically zinc-coated to ASTM B695 Class 55 minimum. Exterior connectors shall follow manufacturer recommendations, or ASTM A653 Type G185	Fasteners and connectors in preservative-treated wood	Exception: plain carbon steel fasteners are permitted in SBX/DOT and zinc borate treated wood in an interior, dry environment	Conditional	Specify hot-dip, stainless, bronze or copper (or G185 connectors) unless the interior dry SBX/DOT exception is documented	Verified
2304.10.6.3	Exterior or wet FRTW fasteners	Fasteners for FRTW in exterior, wet or damp locations shall be hot-dipped zinc-coated galvanized steel, stainless steel, silicon bronze or copper. Staples shall be stainless. Other fasteners may be mechanically zinc-coated to ASTM B695 Class 55 minimum	FRTW in exterior, wet or damp locations	Interior FRTW fasteners follow 2304.10.6.4	Conditional	Use the same corrosion-resistant set as exterior PT wood on wet-location FRTW	Verified
2304.10.6.4	Interior FRTW fasteners	Fasteners for interior FRTW shall follow the manufacturer's recommendations; in their absence, 2304.10.6.3 applies	Interior FRTW	Manufacturer letter overrides the default exterior set	Direct	File the FRTW fastener letter; if none, specify the 2304.10.6.3 metals	Verified
2304.10.7	Load-path strap thickness	Where wall framing members are not continuous from foundation sill to roof, they shall be secured for a continuous load path. Sheet metal clamps, ties or clips shall be galvanized steel or other approved corrosion-resistant material not less than 0.85 mm base-metal thickness	Wood wall framing that is not continuous storey-to-storey	Applies to wood-frame load path, not to Type I concrete/steel frame	Conditional	If wood stud walls are used, specify strap steel ≥ 0.85 mm ; do not apply this to the tower concrete frame	Verified

8. Decay, termites and balcony ventilation

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2304.12.1.1	Crawl-space wood clearance	Wood joists or wood structural floors closer than 450 mm , or wood girders closer than 300 mm , to exposed ground in crawl spaces or unexcavated areas within the foundation perimeter shall be naturally durable or preservative-treated	Wood floor members near exposed ground	Naturally durable or PT wood is the alternative to the clearance	Conditional	If a crawl space exists, keep joists ≥ 450 mm and girders ≥ 300 mm off earth, or treat the wood	Verified
2304.12.1.2	Foundation-wall wood clearance	Wood framing members, including sheathing, in contact with exterior foundation walls and less than 200 mm from exposed earth shall be naturally durable or preservative-treated	Wood against exterior foundation walls	None stated	Conditional	Keep wood ≥ 200 mm above earth at foundation walls, or specify durable/PT wood	Verified
2304.12.1.3–2304.12.1.4	Below-grade furring and sleepers	Wood framing and furring in direct contact with the interior of exterior masonry or concrete walls below grade , and sleepers and sills on a concrete or masonry slab in direct contact with earth , shall be naturally durable or preservative-treated	Basement furring or sleepers on earth-supported slabs	Interior slabs isolated by an intact moisture barrier are not “direct contact with earth” (commentary is not adopted as a numbered exception; confirm the published code if relying on a barrier)	Conditional	Specify PT or naturally durable sleepers and below-grade furring	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2304.12.1.5	Wood siding grade clearance	Clearance between wood siding and earth shall be not less than 150 mm , and not less than 50 mm vertical from concrete steps, porch slabs, patio slabs and similar weather-exposed horizontal surfaces, except where siding, sheathing and wall framing are naturally durable or preservative-treated	Exterior wood siding	Durable/PT siding, sheathing and framing together waive the clearance	Conditional	If wood siding is used, hold it ≥ 150 mm off earth and ≥ 50 mm off weather-exposed slabs	Verified
2304.12.2	Interior PT sealer	Preservative-treated wood used in interior locations shall be protected with two coats of urethane, shellac, latex epoxy or varnish unless waterborne preservatives are used, after drying per the manufacturer	Interior PT wood that is not waterborne-treated	Waterborne PT wood does not take the two-coat sealer	Conditional	On solvent-treated interior PT, specify two coats of the listed sealer	Verified
2304.12.2.1	Girder-end airspace	Ends of wood girders entering exterior masonry or concrete walls shall have a 12.5 mm airspace on top, sides and end, unless naturally durable or preservative-treated wood is used	Wood girders bearing into exterior masonry or concrete	Durable/PT wood waives the airspace	Conditional	Detail a 12.5 mm air gap at untreated girder pockets	Verified
2304.12.2.2	Posts on earth-supported slabs	Posts or columns supporting permanent structures and supported by a concrete or masonry slab or footing in direct contact with the earth shall be naturally durable or preservative-treated	Wood posts on earth-supported pads	Exception: not weather-exposed (or roof-protected), on piers/pedestals projecting not less than 25 mm with an impervious moisture barrier, and not less than 200 mm above exposed earth	Conditional	Treat wood posts at grade, or isolate them 25 mm above the pad and 200 mm above earth	Verified
2304.12.2.3	Exposed balcony supports	Naturally durable or preservative-treated wood shall be used for portions of wood members that form structural supports of buildings, balconies, porches or similar appurtenances where exposed to the weather without adequate roof/eave protection against moisture	Weather-exposed wood balcony or similar supports	Exception: sawn lumber in a region where experience shows climate precludes the need for durable materials — AHJ confirmation required	Conditional	If wood balcony structure is used, specify naturally durable or PT wood unless the AHJ accepts the dry-climate exception	Verified
2304.12.2.4	Wood under permeable weather slabs	Wood structural members supporting moisture-permeable floors or roofs exposed to the weather (such as concrete or masonry slabs) shall be naturally durable or preservative-treated unless separated by an impervious moisture barrier that provides positive drainage of infiltrating water	Wood supporting weather-exposed concrete or masonry topping	Impervious barrier with positive drainage is the alternative	Conditional	If a wood-supported terrace exists, either treat the wood or detail a drained moisture barrier	Verified
2304.12.2.5	Enclosed balcony ventilation	Enclosed framing in exterior balconies and elevated walking surfaces that have weather-exposed surfaces shall have openings providing a net free cross-ventilation area not less than 1/150 of the area of each separate space	Enclosed wood (or enclosed) framing in weather-exposed balconies / elevated walks	Applies to enclosed framing; open-joint balconies are a different condition	Conditional	If balcony structure is enclosed, provide vents ≥ 1/150 of each cavity	Verified
2304.12.2.6–2304.12.2.6.1	Ground-contact wood	Wood in contact with exposed earth shall be naturally durable for decay and termites or preservative-treated to AWP A U1 for soil or freshwater use. Posts and columns supporting permanent structures and embedded in concrete exposed to weather or in direct contact with the earth shall be preservative-treated	Wood in earth or embedded in weather-exposed / earth-contact concrete	Exception: untreated wood continuously and entirely below ground-water level or submerged in fresh water	Conditional	Do not embed untreated wood posts in grade-level concrete	Verified
2304.12.2.7	Very-heavy termite areas	In geographical areas where termite hazard is known to be very heavy , wood floor framing in 2304.12.2.1 locations and exposed framing of exterior decks or balconies shall be naturally durable (termite-resistant) or preservative-treated to AWP A U1, or provided with approved termite protection	AHJ-designated very-heavy termite area	Not a default Riyadh rule in this extract	External verification	Confirm termite designation with the AHJ before leaving balcony or floor wood untreated	Verified
2304.12.3–2304.12.4	Attic and crawl-space ventilation pointer	For attic ventilation see 1202.2.2 . For under-floor (crawl space) ventilation see 1202.4	Attics or crawl spaces	Ventilation ratios are not in this chapter	External verification	Take attic and crawl-space free-area from Chapter 12; do not invent ratios here	Verified
2304.13	Wood supporting masonry long-term	Wood members supporting concrete, masonry or similar materials shall be checked for long-term loading using ANSI/AWC NDS, with total deflection limited per 1604.3.1	Wood supporting concrete or masonry	Exception: horizontal wood supporting masonry or concrete nonstructural floor or roof surfacing not more than 100 mm thick need not be checked for long-term loading	Conditional	If wood supports a topping thicker than 100 mm , require an NDS creep check; do not import 1604.3.1 limits here	Verified

9. Interior paneling

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2304.7	Interior wood paneling standards	Softwood wood structural panels used for interior paneling shall conform to Chapter 8 and be installed in accordance with Table 2304.10.2 . Panels shall comply with DOC PS 1, DOC PS 2 or ANSI/APA PRP 210. Prefinished hardwood paneling shall meet ANSI A135.5 . Hardwood plywood shall conform to HPVA HP-1	Interior wood paneling	Chapter 8 flame-spread classes and Table 2304.10.2 nail schedules are not imported (the fastening table is flattened OCR)	External verification	Send interior paneling to Chapter 8 for class and to a published Table 2304.10.2 for nailing; do not reconstruct either here	Verified

10. Heavy timber and CLT sizes (Type IV)

Use these rows only if the gap register opens Type IV / mass timber. Table 2304.11 body cells are not adopted.

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2304.11	Heavy-timber charging	Where a structure, portion or element is required to be heavy timber, comply with 2304.11.1 through 2304.11.4. Minimum dimensions shall comply with Table 2304.11 or with 2304.11.2 through 2304.11.4. Lumber decking shall be in accordance with 2304.9	Code-required heavy timber (typically Type IV)	Table 2304.11 glued-lam / SCL columns are concatenated OCR and are not adopted. Running-text sizes in 2304.11.2–2304.11.4 remain usable	Conditional	If Type IV is selected, size walls/floors/roofs from 2304.11.2–11.4 and verify the published Table 2304.11 for columns, beams and arches	Verify source
2304.10.1	Type IV connection fire rating	Fire-resistance ratings for connections in Type IV-A, IV-B or IV-C shall be by testing per 703.2 (connection part of the test) or by engineering analysis limiting average temperature rise to 140°C and maximum rise to 180°C for the required rating time	Type IV-A / IV-B / IV-C connections	Analysis includes connectors, fasteners and portions of wood members in the structural design of the connection	Conditional	If mass timber is used, show connection protection by listing or by the 140°C / 180°C analysis	Verified
2304.11.1.1–2304.11.1.3	Heavy-timber continuity	Columns shall be continuous or superimposed through all stories and connected in an approved manner. Wood bolsters shall not sit on column tops unless the columns support roof loads only . Floor beams on masonry/concrete shall use approved wall-plate boxes or hangers. Every roof girder and not less than every alternate roof beam shall be anchored to resist Chapter 16 forces	Heavy-timber columns, floors and roofs	Chapter 16 anchorage forces are not imported	Conditional	Detail continuous columns, hangers at masonry, and roof-beam anchorage; do not copy wind/seismic forces from this chapter	Verified
2304.11.2.1	CLT exterior wall thickness	Exterior walls may be cross-laminated timber not less than 100 mm in thickness meeting 2303.1.4	Type IV / heavy-timber exterior walls of CLT	PRG 320 identity is 2303.1.4	Conditional	Specify CLT exterior walls ≥ 100 mm actual if this wall type is used	Verified
2304.11.2.2	Heavy-timber interior walls	Interior walls and partitions shall be solid wood of not less than two layers of 25 mm matched boards, or laminated construction 100 mm thick, or 1-hour fire-resistance-rated construction	Heavy-timber interior walls and partitions	The 1-hour option is not a wood-thickness substitute for a higher required rating elsewhere	Conditional	Choose two 25 mm matched layers, 100 mm laminated, or a listed 1-hour assembly	Verified
2304.11.3–2304.11.3.1	CLT floor thickness	Floors shall be without concealed spaces or with concealed spaces complying with 602.4.4.3 . CLT floors shall be not less than 100 mm actual thickness, continuous support-to-support and mechanically fastened	Heavy-timber / Type IV floors of CLT	602.4.4.3 concealed-space rules are not imported	Conditional	Specify CLT floors ≥ 100 mm actual and either no cavities or 602.4.4.3 detailing	Verified
2304.11.3.2	Sawn or glulam plank floors	Option 1: splined or T&G planks not less than 75 mm nominal, covered with 25 mm nominal T&G flooring laid crosswise or diagonally, 12.5 mm wood structural panel, or 12.5 mm particleboard. Option 2: planks not less than 100 mm nominal in width set on edge, well spiked, with the same overlay thicknesses. Continuous joints only at supports. Floors shall not extend closer than 12.5 mm to walls; that gap shall be covered by wall-fixed moulding that does not restrain shrinkage	Heavy-timber sawn or glulam plank floors	Masonry corbelling may replace the moulding	Conditional	Pick option 1 or 2; keep a 12.5 mm shrinkage gap at walls with wall-fixed cover	Verified
2304.11.4–2304.11.4.1	CLT roof thickness	Roofs shall be without concealed spaces or with concealed spaces complying with 602.4.4.3 . CLT roofs shall be not less than 75 mm nominal in thickness, continuous support-to-support and mechanically fastened. Wall-supported roof decks shall be anchored for Chapter 16 forces with steel bolts, lags, screws or approved hardware	Heavy-timber / Type IV CLT roofs	Chapter 16 forces are not imported	Conditional	Specify CLT roofs ≥ 75 mm nominal and show steel anchorage to walls	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2304.11.4.2	Sawn, panel or glulam plank roofs	Sawn or glulam splined or T&G plank not less than 50 mm nominal; or 32 mm wood structural panel (exterior glue); or planks not less than 75 mm nominal in width set on edge as required for floors	Heavy-timber sawn / WSP / glulam roofs	Equivalent fire-resistance and structural properties may be submitted as an alternative	Conditional	Specify 50 mm T&G, 32 mm exterior-glue WSP, or 75 mm on-edge planks	Verified
Table 2304.11 note b	Sprinklered roof-member width	Where protected by approved automatic sprinklers under the roof deck, framing members shall be not less than 75 mm nominal in width	Heavy-timber roof framing under a sprinklered deck	Note a (spaced members 75 mm / cover plate 50 mm) is also in the extract. Table body sizes remain Verify source	Conditional	If Type IV roof framing uses the sprinkler reduction, confirm the published note and keep width $\geq$ 75 mm	Verified

11. Project-use controls

1. Use **Verified** rows for initial scoping after the row trigger is confirmed (the wood product or Type IV system actually exists).
2. Do not use **Verify source** rows, including reconstructed fastening, sheathing, diaphragm, shear-wall or Table 2304.11 body cells, as design-release values.
3. Do not import Section 603.1 FRTW hours, storey limits or the **6 m** roof clearance into this matrix; those live in Chapter 6.
4. Do not import Chapter 8 interior-finish classes, Chapter 16 loads, Chapter 12 ventilation ratios, NDS/SDPWS stresses, or WFCM dwelling tables.
5. Do not treat General Comments (only Types III/IV/V as wood buildings) or commentary examples as numbered code cells.
6. Do not use Section 2308 as a KSA prescriptive wood-frame path.
7. Omit wood-frame fastening, sheathing-wind, diaphragm and shear-wall tables from issued architectural drawings unless the gap register opens a wood podium or Type IV LFRS.
8. Record construction-type, FRTW listing and wood-product decisions in the project Golden Thread; this matrix is not evidence of SCD NOC or stamped compliance.

12. Coverage summary

Internal inventory of the attached Chapter 23 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)
- **Total independently checkable numeric records:** 286
- **Verified:** 148
- **Verify source:** 138

Counts by top-level section

Top-level section	Records
2301	0
2302	0
2303	22
2304	180
2305	25
2306	59
2307	0
2308	0
2309	0

Appended tables in this extract: 2304.6.1, 2304.8(1)–(5), 2304.9.3.2, 2304.10.1 / 2304.10.2, 2304.11, 2305.2(1)–(2), 2306.1, 2306.1.4, 2306.2(1)–(2), 2306.3(1)–(3). Table 2306.1 is a standards list (no numeric records). Remaining table bodies are concatenated OCR and are counted as **Verify source**. Readable footnotes (including Table 2304.11 notes a and b, Table 2304.8 deflection tokens, and Table 2306 duration factors **0.63** / **0.56**) are counted as **Verified** and are not design-release values for omitted wood-frame systems.

Coverage cross-check against SBC 201 Chapter 23 Woods (2024)_CS.md was topics-only: chapter scope versus Chapter 6; FRTW identity (2303.2); Type I/II pointer to 603.1; heavy-timber / CLT running-text sizes; Table 2304.11 OCR warning. No CS.md value was copied into a matrix cell.

Wood-frame fastening, sheathing, decking, diaphragm and shear-wall numeric rules are included in the inventory counts and omitted from project-use as not typical for the Type I R-2 high-rise shaft, except where the gap register opened Type IV.

13. Unresolved-source register

Affected table / clause	Why unverified	Control note
2303.6 staple bending (No. 16 gage)	Code text reads “ 3.6 0.40 N·m ” — concatenated OCR. No. 15 (0.45 N·m) and No. 14 (0.50 N·m) are readable and were inventoried as Verified but omitted from lead tables as not typical	Do not repair the No. 16 value from memory. Nail yield strengths 550 / 620 / 690 MPa in the same paragraph are unambiguous
Table 2304.6.1	Concatenated wall-sheathing wind-speed / nail / span-rating cells. Charging 2304.6.1 also limits the table to mean roof height not greater than 9 m	Do not reconstruct Vasd cells. The table is not applicable to a high-rise shaft
Tables 2304.8(1)–(5)	Flattened span, thickness and load cells. Selected footnotes (19-percent, 1/180, 1/240, 6.3 mm, 37.5 mm, 4.80 / 3.10 kN/m², 0.72 kN/m²) are readable	Use only a published table for wood floor/roof sheathing; not typical for Type I floors
Table 2304.9.3.2	Concatenated laminated-decking nail size / spacing rows	Do not adopt reconstructed nailing; lumber decking is not typical for this tower
Tables 2304.10.1 / 2304.10.2	Fastening schedule OCR loops, mixed headers (10.1 vs 10.2) and duplicated 16d strings	Do not reconstruct nail counts. 2304.7 installation pointer remains outbound to the published table

Affected table / clause	Why unverified	Control note
Table 2304.11 body	Glued-laminated / SCL width-depth cells concatenated. Running text 2304.11.2–2304.11.4 is usable. Notes a and b (75 mm / 50 mm spaced members; sprinklered roof width 75 mm) are readable	Verify the published table before sizing Type IV columns, beams, girders or arches
Tables 2305.2(1)–(2)	Fastener-slip en and panel Gt cells concatenated	Do not use reconstructed en / Gt in Equations 23-1 / 23-2
Table 2306.1.4	Decking moment/deflection formulae concatenated	Do not compute decking capacity from the OCR image
Tables 2306.2(1)–(2) and 2306.3(1)–(3)	Staple diaphragm and shear-wall allowable-shear cells concatenated. Readable footnotes include species factors 0.82 / 0.65 , duration 0.63 / 0.56, 40 percent wind increase in the charging text, Type V-only Table 2306.3(2), and SDC D/E/F 5110 N/m framing	Do not reconstruct staple shears. Wood-frame LFRS is not typical for the Type I shaft